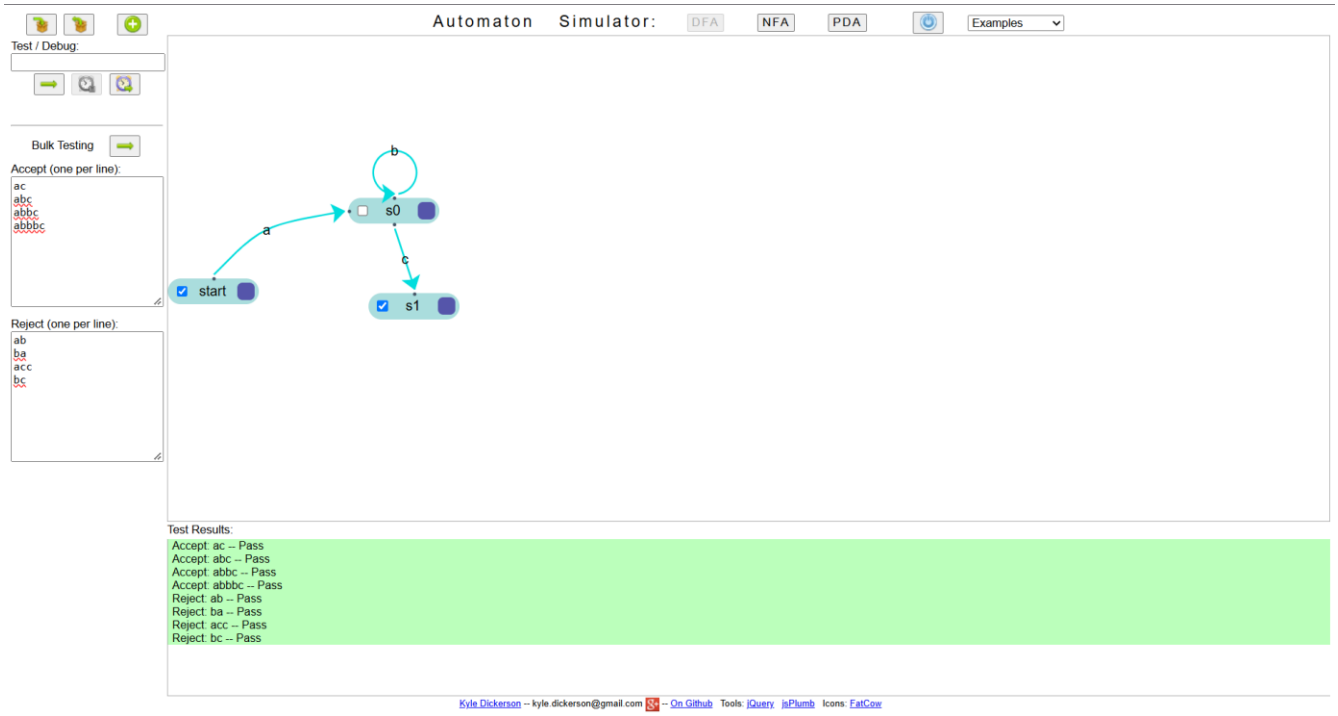
1. A company is developing a smart security system for a research laboratory. Every employee entering the lab uses a digital access card that generates a sequence of symbols:

- a → Valid ID scan
- b → Biometric verification
- c → Door access confirmation

The security rule states that:

1. Every valid access sequence must start with a.
2. It may contain any number of biometric checks (b).
3. The sequence must end with c.

Examples of valid sequences: ac, abc, abbc, abbbc



2.Given a DFA used for validating a specific PIN pattern bbbc, bca ,c, ca, caaaaaa minimize the DFA and explain how state reduction improves performance and efficiency in embedded systems

